

Figure 7(b). TERA layers x cross-cutting concerns: per-cell prescription.

	C1 Security	C2 Evaluation	C3 Observability	C4 Governance	C5 LLMOps
L0 Cloud foundation	per-tenant cache KMS keys net deny-default		OTel GenAI correlation IDs	append-only audit SBOM AICM mapping	GitOps eval-gated promo rollback runbook
L1 Data and knowledge	source allow-list encrypted embeddings	retrieval-quality SLO	ingest telemetry	provenance graph PII scan log	embedding versioning
L2 Retrieval	rate-limit multi-source order check	RAGAS / ARES SYNCHECK CI gate	retrieval traces	retrieval-quality evidence	retriever versioning
L3 Reasoning and planning	StruQ queries PromptShield multi-turn safety	agent-capability benchmarks	plan-trace span	plan log retention	agent versioning
L4 Tool and action	hash-pinned tools ACLs ToolHijacker mit.	BFCL tool-use correctness	tool-call traces side-effect logs	HITL approval gates; reversal	tool registry versioning
L5 Application	output safety classifier collusion canary	outcome metrics	user-facing telemetry	user disclosure audit	deploy / rollback cost SLO